Project Report: Advanced Multi-Cycle Branch Predictor — RTL to GDSII

Overview

This project implements a 4-bit GHR (Global History Register) two-level adaptive branch predictor with a 3-stage multi-cycle pipeline, taken through a complete RTL-to-GDSII physical design flow using the Cadence ASIC toolchain (Genus for synthesis, Innovus for physical implementation) on a 90nm process.

Technologies Used

Hardware Description & Simulation - Verilog (RTL design) - Icarus Verilog (iverilog) — open-source functional simulation cross-check - Cadence NC-Launch / SimVision — functional simulation and waveform analysis

Cadence Toolchain (Physical Design) - Cadence Genus — logic synthesis - Cadence Innovus — floorplanning, power planning, placement, clock tree synthesis, routing, timing signoff

Supporting Tools - Tcl scripting for synthesis/place-and-route automation - Linux/bash environment

Key Achievements

- Delivered a fully clean RTL-to-GDSII implementation — zero DRC violations, clean netlist-to-layout match, and closed timing (met setup slack of 6508 ps on a 10 ns clock period).
- Independently diagnosed and fixed genuine RTL synthesizability issues rather than working around them, verifying functional correctness before and after every change.
- Verified functional correctness across 150 simulated branches, achieving 88% prediction accuracy (133 correct, 17 flushes), with results cross-checked using both Cadence SimVision and open-source Icarus Verilog.
- Gained hands-on, end-to-end exposure to the full ASIC design flow: simulation → synthesis → floorplanning → power planning → placement → clock tree synthesis → routing → timing signoff → GDSII export.

Major Challenges Faced & Resolutions

1. Synthesis elaboration failure — combined async reset/flush condition

Challenge: Cadence Genus failed to elaborate the design with a `CDFG-364` “Unsynthesizable Process” error. **Root cause:** The `multi_cycle_pipeline` module’s clocked always block combined `if (reset || flush)` inside a block sensitive to `posedge clk or posedge reset` — mixing an async-listed signal (`reset`) with a signal not in the sensitivity list (`flush`) in the same top-level branch condition. **Resolution:** Split the condition into `if (reset) ... else if (flush) ...` — functionally identical (both branches clear all registers to zero) but structurally unambiguous for the synthesis tool.

2. Silently ignored register initializer

Challenge: Genus issued a `VLOGPT-37` warning: "Ignoring unsynthesizable construct… reg declaration with initial value," meaning the `pattern1` register's inline initial value would never actually be applied in synthesized hardware — despite simulating correctly in RTL. **Resolution:** Since `pattern1`'s value never changes at runtime, it was converted from a `reg` with an inline initializer to a `localparam`, which is evaluated at elaboration time and requires no reset logic.

3. LEF file load order errors in Innovus

Challenge: Innovus repeatedly failed during Design Import with `"Layer 'Metal1' is not defined"` and `"No technology information is defined in the first LEF file"`. **Root cause:** The macro LEF (defining standard cell geometry) was being loaded before the technology LEF (defining the metal layer stack), so layer references in the macro file could not resolve. This was further complicated by a technology LEF that, per its own header comments, only defined via geometry — not the base metal layer stack. **Resolution:** Identified the correct technology LEF file and enforced the correct load order: technology LEF first, macro LEF second.

4. Post-route optimization blocked by missing On-Chip Variation setting

Challenge: `optDesign` failed post-route with error `IMPOPT-7027`, refusing to proceed without On-Chip Variation (OCV) analysis mode explicitly enabled. **Resolution:** Ran `setAnalysisMode -analysisType onChipVariation -cppr both`, enabling variation-aware timing analysis with clock reconvergence pessimism removal for both setup and hold checks — a required step for accurate post-route signoff timing.

Results Summary

Functional Simulation

Metric	Value
Total Branches Resolved	150
Correct Predictions	133
Pipeline Flushes	17
Prediction Accuracy	88%

Synthesis (Genus, 90nm, slow corner)

Metric	Value
Total Cell Count	696
Total Area	6497.23 μm^2
Total Power	0.729 mW
Worst Setup Slack	6508 ps (MET)

Physical Design (Innovus)

- Power delivery network: Metal9/Metal8 rings and stripes, 0.5 µm width/spacing
- Placement density: 70.5% post-optimization
- Clock tree synthesized via `ccopt_design`
- Post-route timing signed off under OCV analysis with CPPR enabled

Future Improvements

- **Predictor accuracy:** Explore more advanced branch prediction schemes (e.g., gshare, a longer GHR, or a hashed indexing function) to push prediction accuracy meaningfully above the current 88%.
- **Timing optimization:** With significant unused setup slack on a 10 ns clock, re-synthesize targeting a faster clock period to explore the design's true maximum operating frequency.
- **Power optimization:** Investigate clock gating on the pipeline registers and predictor table, since sequential logic dominates total power (~69% per the power report).
- **Area optimization:** Increase core utilization for a denser, more area-efficient floorplan, now that the design is known to route and close timing cleanly at looser settings.
- **Scalability:** Extend the design to a larger GHR (e.g., 8-bit) and evaluate how the current unrolled-register synthesis approach scales, likely requiring a return to proper memory-macro inference for larger prediction tables.
- **Automation:** Convert the manual Innovus GUI-driven flow into a fully scripted, non-interactive Tcl flow for faster, repeatable re-runs.